

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Mathematics — Class IX

Syllabus: Unit 1 — Real Numbers | Unit 2 — Logarithms | Unit 4 — Factorization & Algebraic Manipulation | Unit 5 — Linear Equations & Inequalities | Unit 7 — Coordinate Geometry

Total Marks: 65

Time Allowed: 2 Hours 30 Minutes

Version: 2

General Instructions:

1. Attempt all questions from Section A (MCQs). Bubble the correct answer in the Answer column.
2. Attempt all 11 questions from Section B. For each question, attempt either the main part OR its alternative — not both.
3. Attempt all 4 questions from Section C. For each question, attempt either the main part OR its alternative — not both.
4. Write legibly. Show all working clearly.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	Which property is used in: $a^m \div a^n = a^{m-n}$?	☐ Product rule	☐ Quotient rule	☐ Power rule	☐ Neg. exponent	
2	If $a > 0$, the expression $\sqrt{a}$ is always:	☐ negative	☐ irrational	☐ non-negative real	☐ integer	
3	Scientific notation of 0.000307 is:	☐ 3.07×10^4	☐ 3.07×10^{-4}	☐ 3.07×10^{-3}	☐ 30.7×10^{-5}	
4	$\log_3 81$ equals:	☐ 3	☐ 27	☐ 4	☐ 9	
5	$\log 18$ expressed using $\log 2$ and $\log 3$ is:	☐ $\log 2 + \log 3$	☐ $\log 2 + 2 \log 3$	☐ $2 \log 2 + \log 3$	☐ $3 \log 2 + \log 3$	
6	Factorization of $x^2 - 9x + 20$ is:	☐ $(x-4)(x-5)$	☐ $(x+4)(x+5)$	☐ $(x-4)(x+5)$	☐ $(x+4)(x-5)$	
7	For polynomials P and Q: HCF × LCM equals:	☐ $P + Q$	☐ $P - Q$	☐ $P \times Q$	☐ $P \div Q$	
8	Standard form of a linear equation in one variable is:	☐ $ax^2 + b = 0$	☐ $ax + b = 0$	☐ $ax + by = 0$	☐ $a + b = 0$	
9	Solution set of $ 3x - 6 = 9$ is:	☐ $\{5, -1\}$	☐ $\{5, 1\}$	☐ $\{-1, -5\}$	☐ $\{3, -3\}$	
10	Distance between (0, 0) and (5, 12) is:	☐ 17	☐ $\sqrt{119}$	☐ 13	☐ 7	
11	Midpoint of A(4, -2) and B(-2, 8) is:	☐ (2, 6)	☐ (1, 3)	☐ (3, 3)	☐ (1, 5)	
12	The HCF of $x^2 - 4$ and $x^2 - x - 2$ is:	☐ $(x+2)$	☐ $(x-2)$	☐ $(x^2 - 4)$	☐ $(x-1)(x+2)$	

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt all 11 questions. Answer the main part OR its alternative for each.

Part	Question	Mks	OR	Alternative Question	Mks
2(i)	Simplify using laws of exponents: $(-3)^2 \cdot (-3)^{-4} \cdot (-3)^3 \div (-3)^{-2}$	3	OR	Write as equivalent radical and simplify: (i) $(-8)^{1/3}$ (ii) $(\sqrt{25})^4$ (iii) $(x-3)^2 \cdot (x^0)^5$	3
2(ii)	State TRUE or FALSE; if false give counterexample: (a) Every integer is rational. (b) Every irrational is real. (c) Every rational is an integer.	3	OR	Represent on the number line: (i) $-3\frac{1}{2}$ (ii) $19/4$ (iii) $x \geq -2$	3
2(iii)	Simplify; give answer in scientific notation : $(4.2 \times 10^5) \times (3.0 \times 10^{-3}) \div (6.0 \times 10^4)$	3	OR	Convert: (i) 7.05×10^{-4} to standard notation (ii) 2.31×10^6 to standard notation (iii) 0.0000000512 to scientific notation	3
2(iv)	Find x: (i) $\log_x 125 = 3$ (ii) $\log_4 x = -3/2$ (iii) $\log_2(x^2-3) = \log_2 13$	3	OR	Expand: $\log \sqrt{(53.3/46.4)}$ Then combine into single term: $3 \log p - 2 \log q + (1/2) \log r$	3
2(v)	Find log using tables: (i) 4580 (ii) 0.00093 (iii) 54	3	OR	Find antilog of: (i) 2.4324 (ii) $\bar{3}.5219$ (iii) 0.0000	3
2(vi)	Factorize completely: (i) $x^4 + 4m^4$ (ii) $3x^2 + 11x + 6$	3	OR	Factorize: (i) $8a^3 - 125b^3$ (ii) $27p^3 - 108p^2q + 144pq^2 - 64q^3$	3
2(vii)	Find HCF by factorization: $x^2 - 49$ and $x^2 - 4x - 21$	3	OR	Find LCM of: $x^2 + x - 6$, $x^2 + x - 2$, $x^2 - 4x + 3$	3
2(viii)	Solve for x: $(2x+3)/5 = (3-4x)/8$	3	OR	Solve (write ϕ if extraneous): $\sqrt{(5x-4)} = \sqrt{(7x+2)}$	3
2(ix)	Solve where $x \in \mathbb{R}$: $ 4x - 3 = 13$	3	OR	Solve compound inequality ($x \in \mathbb{R}$): $3x+2 \leq 11$ and $2x-5 > -3$	3
2(x)	Use distance formula to check if A(1,-2), B(7,8) and C(4,3) are collinear .	3	OR	Prove that A(-4,2), B(1,4), C(3,-1) and D(-2,-3) are vertices of a square .	3
2(xi)	P(3,k) is midpoint of A(-1,7) and B(7,-3). Find k, then find distance PA.	3	OR	Points (2,3) and (4,5) are at equal distance from (3,t). Find t.	3

SECTION C — Long Questions (4 × 5 = 20 Marks)

Part	Question	Mks	OR	Alternative Question	Mks
3(i)	(a) Given $\log_b 2 = 0.3010$, $\log_b 3 = 0.4771$, $\log_b 5 = 0.6990$, evaluate: (i) $\log_b 30$ (ii) $\log_b(100/9)$ (iii) $\log_b \sqrt{(3\sqrt{450})}$ (b) Find number of digits in 3^{30} .	5	OR	Evaluate using log tables: $x = \sqrt[3]{(8.59) \times (55.6)^2 \div (2.51 \times \sqrt{2.12})}$ Show: log x expansion, table values, antilog x.	5
3(ii)	Simplify: (a) $[(x^2-1)/(x^2+4x+4)] \times [(x+2)/(x^2+2x-3)]$ (b) $1/[x(x-y)] + 1/[y(x+y)]$	5	OR	(a) Find square root of $4a^2 + 12ab + 9b^2$ by factorization. (b) Factorize: (i) $(x^2+3x-4)(x^2+3x+5)+8$ (ii) $8a^3 - 125b^3 - 2a + 5b$	5
3(iii)	(a) Solve ($y \in \mathbb{R}$): $(y+1)/3 + (y+1)/2 = 2 - (y+3)/2$ (b) Solve and write in set notation: $(3/4)x - 1 \geq x + 1$, $x \in \mathbb{R}$	5	OR	(a) Solve: $ 10-x /5 = 2x-5 /2$, $x \in \mathbb{R}$ (b) The sum of 2x and 3 is between 5 and 17. Between what two values does x lie? (Set notation)	5
3(iv)	Vertices A(0,0), B(6,0), C(6,4), D(0,4) form a quadrilateral. (a) Find AB, BC, CD, DA and show sides are NOT all equal. (b) Find lengths of diagonals AC and BD, and the midpoints of both diagonals. Hence prove ABCD is a rectangle .	5	OR	Triangle PQR has vertices P(1,2), Q(7,2), R(4,8). (a) Find PQ, PR, QR. Show the triangle is isosceles . (b) Find midpoint M of base PQ. Show $RM \perp PQ$ using slopes.	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Unit 1 (Real Numbers) | Unit 2 (Logarithms) | Unit 4 (Factorization & Algebraic Manipulation) | Unit 5 (Linear Equations & Inequalities) | Unit 7 (Coordinate Geometry)

— End of Question Paper —